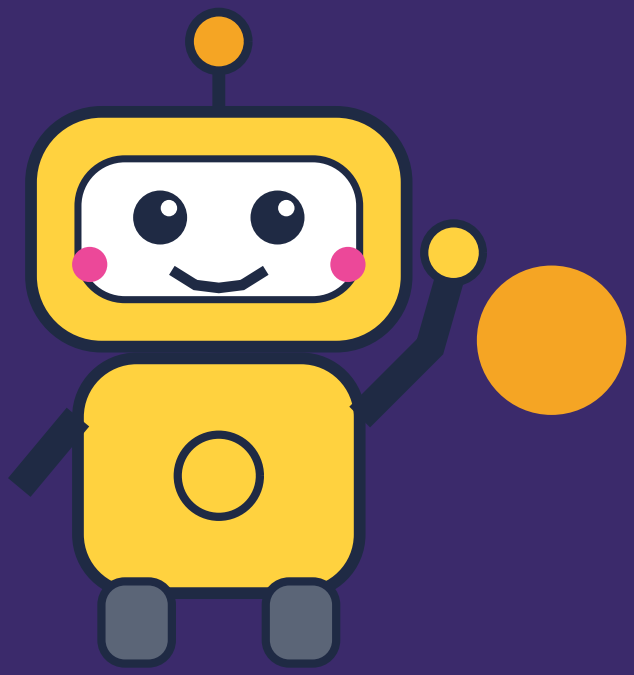
ASSESSMENT BOOK

Think Like a Coder

Chapter assessments for Class 8

11 one-page assessments, one for every chapter
10 marks each: choose, apply and think like a coder
Answers, marking guides and a class record sheet



For CBSE's Computational Thinking and AI curriculum, 2026-27

Goes with Think Like a Coder, Class 8 · Photocopiable for classroom use

SAMPLE EDITION · DRAFT

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Using these assessments

This book has one assessment for every chapter of *Think Like a Coder*, Class 8. Each assessment fits on one page, takes about 30 minutes and is marked out of 10. Use it after the chapter's last period, or as a revision check later in the term. All questions are new: they test the same skills as the workbook, not the same questions.

Section	What it checks	Questions	Marks
A	Knowing and understanding the chapter's ideas	3 multiple choice	3 x 1
B	Applying the idea to a short problem, with working	2 short answers	2 x 2
C	Thinking like a coder: write, trace, fix or explain a program or plan	1 task	1 x 3

Turning a score into a journal level

Score	Journal level	What it usually means
9-10	Extending	Solves new problems and explains the reasoning
7-8	Secure	Applies the idea correctly on their own
4-6	Developing	Understands with prompts; makes some slips
0-3	Beginning	Needs more practice with the chapter's idea

✓ Good practice

- ✓ Read the questions aloud for children who find reading hard: we are testing thinking, not reading.
- ✓ Give full marks for any correct method, even if it differs from the answer key; the marking guides list what to look for.
- ✓ Give part marks in Sections B and C for correct working, even when the final answer has a slip.
- ✓ Record the score on the class record sheet, and use it with your observations in the journal, not instead of them.
- ✓ Go over the test with the class: the most common mistakes are a lesson plan for the next period.

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Assessment 1 A Square and a Cube

Coder skill: Test squares and cubes with prime factors and loops · Level C6 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 Which number is a perfect square? [1 mark]

a) 50 **b)** 81 **c)** 90

2 Which number is a perfect cube? [1 mark]

a) 27 **b)** 36 **c)** 49

3 A number is a perfect square when its prime factors ... [1 mark]

a) pair up **b)** make groups of three

c) are all 2

Section B Answer in a few words or steps — 4 marks

4 Is 324 a perfect square? Use prime factors. [2 marks]

5 What is the smallest number you must multiply 72 by to get a perfect cube? [2 marks]

Section C Think like a coder — 3 marks

6 Algorithm: set k to 1; repeat while k x k < n: k = k + 1; then if k x k = n say 'square' else say 'not a square'. Trace it for n = 50. [3 marks]

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Assessment 2 Power Play

Coder skill: Exponents, doubling and counting combinations · Level C6 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 2^5 = ... [1 mark]

a) 10 **b)** 25 **c)** 32

2 $3^2 \times 3^4$ = ... [1 mark]

a) 3^6 **b)** 3^8

c) 9^6

3 3 shirts and 5 trousers make how many different outfits? [1 mark]

a) 8 **b)** 15 **c)** 35

Section B Answer in a few words or steps — 4 marks

4 Simplify $(5^3 \times 5^4) \div 5^5$. [2 marks]

5 How many 4-digit PINs can be made from the digits 0 to 9? How many 5-digit PINs? [2 marks]

Section C Think like a coder — 3 marks

6 A cell splits into two every 20 minutes. Start with 1 cell. How many cells are there after 3 hours? When does the number first pass 1,000? [3 marks]

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Assessment 3 A Story of Numbers

Coder skill: Number bases: binary, ternary and changing base · Level C6 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 Base 3 uses the digits ... [1 mark]

a) 0 and 1 **b)** 0, 1 and 2 **c)** 0 to 9

2 The place values in base 2 are ... [1 mark]

a) 1, 10, 100, ... **b)** 1, 2, 4, 8, ...

c) 1, 3, 9, 27, ...

3 Roman numerals have no ... [1 mark]

a) zero and place value **b)** letters

c) symbol for ten

Section B Answer in a few words or steps — 4 marks

4 Change 45 into binary by dividing by 2 again and again. [2 marks]

5 Change the base-3 number 2102 into decimal. [2 marks]

Section C Think like a coder — 3 marks

6 Write 100 in base 3 by repeated division. Then check your answer. [3 marks]

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Assessment 4 Quadrilaterals

Coder skill: Classify shapes with if-else rules · Level C6 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 The angles of a quadrilateral add up to ... [1 mark]

a) 180° **b)** 360° **c)** 540°

2 A parallelogram with all sides equal is a ... [1 mark]

a) rectangle **b)** rhombus **c)** kite

3 A quadrilateral with exactly one pair of parallel sides is a ... [1 mark]

a) trapezium **b)** parallelogram **c)** square

Section B Answer in a few words or steps — 4 marks

4 Three angles of a quadrilateral are 85°, 95° and 110°. Find the fourth. [2 marks]

5 True or false? (a) Every square is a rectangle. (b) Every rectangle is a square. [2 marks]

Section C Think like a coder — 3 marks

6 Write if-else rules that name a parallelogram as a square, rectangle, rhombus or just a parallelogram. Use them on a parallelogram with all sides equal and angles of 60° and 120°. [3 marks]

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Assessment 5 Number Play

Coder skill: Divisibility rules and remainders as quick tests · Level C7 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 Which number is divisible by 9? [1 mark]

a) 3,142 **b)** 5,814 **c)** 2,222

2 To test whether a number is divisible by 4, look at ... [1 mark]

a) its last two digits **b)** its digit sum

c) its first digit

3 What day is it 100 days after a Monday? [1 mark]

a) Monday **b)** Tuesday **c)** Wednesday

Section B Answer in a few words or steps — 4 marks

4 Test 6,138 for divisibility by 3, 9 and 11 without dividing. [2 marks]

5 What is the remainder when 1,000 is divided by 7? What day is it 1,000 days after a Sunday? [2 marks]

Section C Think like a coder — 3 marks

6 Write the steps to test divisibility by 11 with the alternating sum. Trace it for 918,082 and for 52,372. [3 marks]

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Assessment 6 We Distribute Yet Things Multiply

Coder skill: The distributive property and identities for fast calculation · Level C7 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 $7 \times (10 + 3) = \ldots$ [1 mark]

a) 73 **b)** 91 **c)** 70

2 $(a + b)^2 = \ldots$ [1 mark]

a) $a^2 + b^2$ **b)** $a^2 + 2ab + b^2$

c) $2a + 2b$

3 Which identity makes 103×97 quick to work out? [1 mark]

a) $(a + b)(a - b) = a^2 - b^2$ **b)** $(a + b)^2$

c) $a(b + c) = ab + ac$

Section B Answer in a few words or steps — 4 marks

4 Work out 51^2 using $(a + b)^2$. [2 marks]

5 Expand $(x + 4)(x + 3)$. [2 marks]

Section C Think like a coder — 3 marks

6 Use an area model to work out 34×26. Show the four parts. [3 marks]

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Assessment 7 Proportional Reasoning

Coder skill: Ratios, the unitary method and scaling · Level C7 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 12 : 18 in simplest form is ... [1 mark]

a) 2 : 3 **b)** 3 : 2 **c)** 6 : 9

2 5 pens cost ₹60. One pen costs ... [1 mark]

a) ₹10 **b)** ₹12 **c)** ₹300

3 More workers taking fewer days is ... [1 mark]

a) direct proportion **b)** inverse proportion

Section B Answer in a few words or steps — 4 marks

4 Are 4 : 7 and 20 : 35 in proportion? Show how you know. [2 marks]

5 8 workers build a wall in 15 days. How long would 12 workers take? [2 marks]

Section C Think like a coder — 3 marks

6 A recipe for 6 people needs 450 g of flour and 3 eggs. Use the unitary method to find the amounts for 10 people. Is this direct or inverse proportion? [3 marks]

How did I do? Colour a face.

I can finish this test.	I can show my thinking.	I can check my answers.
		

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Assessment AI1 AI Project Lifecycle

Coder skill: Define, collect, train, test and deploy · Level C8 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 The first stage of an AI project is ... [1 mark]

a) define the problem **b)** deploy

c) train a model

2 Removing duplicates and missing values is called ... [1 mark]

a) cleaning the data **b)** deploying

c) testing

3 A model is right on 45 of 50 test examples. Its accuracy is ... [1 mark]

a) 45% **b)** 90% **c)** 95%

Section B Answer in a few words or steps — 4 marks

4 Put the stages in order: train, deploy, define, evaluate, collect. [2 marks]

5 AI or automation? (a) A car wash that runs the same steps for every car. (b) An app that spots crop disease from photos of leaves. [2 marks]

Section C Think like a coder — 3 marks

6 A dataset of past students has a row with a missing mark, a row that appears twice, a mark of 340 out of 100, and a 'favourite colour' column. How would you clean each problem? [3 marks]

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Assessment AI2 AI and Its Applications

Coder skill: AI for people and planet; training no-code models · Level C8 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 AI tools that help screen for tuberculosis read ... [1 mark]

a) chest X-rays **b)** satellite images

c) voices

2 In a no-code tool, the labels you create are called ... [1 mark]

a) classes **b)** pixels **c)** layers

3 Camera traps with AI help to ... [1 mark]

a) count wildlife **b)** forecast rain

c) translate languages

Section B Answer in a few words or steps — 4 marks

4 Put in order: train, test with new examples, create classes, add examples. [2 marks]

5 A waste-sorting model trained only on photos with a white background fails in the playground. Why? How would you fix it? [2 marks]

Section C Think like a coder — 3 marks

6 Plan a no-code model that tells ripe tomatoes from unripe ones. Give the classes, the data you would collect, and how you would test it. [3 marks]

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Assessment AI3 Data and Fairness in AI

Coder skill: Spot unbalanced data and rebalance it · Level C8 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 A group with too few examples in the data is ... [1 mark]

a) over-represented **b)** under-represented

2 Fair training data should be ... [1 mark]

a) representative of every group **b)** as small as possible

c) from one city only

3 To check a model for fairness, test ... [1 mark]

a) each group separately **b)** only the biggest group

c) nothing

Section B Answer in a few words or steps — 4 marks

4 A face detector is trained on 900 photos of light-skinned faces and 100 of dark-skinned faces. What problem may it have, and how would you fix it? [2 marks]

5 A translator turns a gender-neutral sentence into 'He is a doctor'. Why does this happen? [2 marks]

Section C Think like a coder — 3 marks

6 A sport-recommending model learnt from 80 boys and 20 girls, and now recommends cricket to everyone. Explain why this is unfair, how you would rebalance the data, and one feature it should not use. [3 marks]

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Assessment AI4 Ethics and Responsible AI

Coder skill: Privacy, checking facts and human accountability · Level C8 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 Giving permission for your data to be used is called ... [1 mark]

a) consent **b)** piracy **c)** bias

2 A torch app asks to read your contacts. You should ... [1 mark]

a) deny it **b)** allow it

c) share your password

3 Who is accountable for a decision made with AI? [1 mark]

a) people **b)** the machine **c)** nobody

Section B Answer in a few words or steps — 4 marks

4 Before sharing a shocking video, name two checks you would make. [2 marks]

5 A map app asks for your location 'all the time'. What smaller permission would you choose, and why? [2 marks]

Section C Think like a coder — 3 marks

6 A bank uses AI to decide who gets a loan. Write one sentence each on how it should be transparent, fair and accountable. [3 marks]

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Answers and marking guide

Accept any correct method. Part marks are shown in brackets for Sections B and C.

1. A Square and a Cube

1. b) 81
2. a) 27
3. a) pair up
4. 324 = (2 x 2) x (3 x 3) x (3 x 3): the factors pair up, so yes; 324 = 18 x 18 (1 mark for the prime factors, 1 mark for the conclusion)
5. 3: 72 = 2 x 2 x 2 x 3 x 3 needs one more 3; 72 x 3 = 216 = 6 x 6 x 6 (1 mark for the prime factors, 1 mark for 3)
6. k x k goes 1, 4, 9, 16, 25, 36, 49, 64. It stops at k = 8 (64 is not less than 50). 64 is not 50, so 'not a square'. (1 mark for the values, 1 mark for stopping at k = 8, 1 mark for the conclusion)

2. Power Play

1. c) 32
2. a) 3^6
3. b) 15
4. $5^2 = 25$ (1 mark for adding and subtracting exponents, 1 mark for 25)
5. 10,000 and 100,000 (1 mark for each)
6. 3 hours = 9 doublings, so $2^9 = 512$ cells. After 10 doublings (3 hours 20 minutes) there are 1,024. (1 mark for 9 doublings, 1 mark for 512, 1 mark for 3 hours 20 minutes)

3. A Story of Numbers

1. b) 0, 1 and 2
2. b) 1, 2, 4, 8, ...
3. a) zero and place value
4. 45, 22 r1, 11 r0, 5 r1, 2 r1, 1 r0, 0 r1: read backwards 101101 (1 mark for the divisions, 1 mark for 101101)
5. 2 x 27 + 1 x 9 + 0 x 3 + 2 = 65 (1 mark for the place values, 1 mark for 65)
6. 100 ÷ 3 = 33 r1; 33 ÷ 3 = 11 r0; 11 ÷ 3 = 3 r2; 3 ÷ 3 = 1 r0; 1 ÷ 3 = 0 r1. Backwards: 10201. Check: 81 + 18 + 1 = 100. (1 mark for the divisions, 1 mark for 10201, 1 mark for the check)

4. Quadrilaterals

1. b) 360°
2. b) rhombus
3. a) trapezium
4. 70° (2 marks; 1 mark for 360° - 290°)
5. (a) true (b) false (1 mark for each)
6. If all angles are 90° and all sides are equal: square. Else if all angles are 90°: rectangle. Else if all sides are equal: rhombus. Else: parallelogram. The test shape is a rhombus. (1 mark for checking square first, 1 mark for all four names, 1 mark for 'rhombus')

5. Number Play

1. b) 5,814
2. a) its last two digits
3. c) Wednesday
4. Digit sum 18: divisible by 3 and 9. Alternating sum 6 - 1 + 3 - 8 = 0: divisible by 11. (1 mark for 3 and 9, 1 mark for 11)
5. 6 (7 x 142 = 994); Saturday (1 mark for each)
6. Add and subtract the digits in turn (+, -, +, ...); if the result is 0 or a multiple of 11, the number is divisible by 11. 918,082: 9 - 1 + 8 - 0 + 8 - 2 = 22, yes. 52,372: 5 - 2 + 3 - 7 + 2 = 1, no. (1 mark for the steps, 1 mark for each trace)

6. We Distribute Yet Things Multiply

1. b) 91
2. b) $a^2 + 2ab + b^2$
3. a) $(a + b)(a - b) = a^2 - b^2$
4. 2,500 + 100 + 1 = 2,601 (1 mark for the identity, 1 mark for 2,601)
5. $x^2 + 7x + 12$ (2 marks; 1 mark for four correct parts before adding)
6. 30 + 4 and 20 + 6: 600, 180, 80 and 24; total 884 (1 mark for the split, 1 mark for the four parts, 1 mark for 884)

7. Proportional Reasoning

1. a) 2 : 3
2. b) ₹12
3. b) inverse proportion
4. Yes: 4 x 35 = 140 and 7 x 20 = 140 (1 mark for the check, 1 mark for yes)
5. 10 days (120 worker-days ÷ 12) (1 mark for 120 worker-days, 1 mark for 10)
6. 75 g per person, so 750 g; half an egg per person, so 5 eggs. Direct proportion. (1 mark for each)

AI 1. AI Project Lifecycle

1. a) define the problem
2. a) cleaning the data
3. b) 90%
4. define, collect, train, evaluate, deploy (2 marks; 1 mark if one stage is out of place)
5. (a) automation (b) AI (1 mark for each)
6. Fill or drop the missing value; delete the duplicate row; correct or remove the impossible mark; drop the irrelevant column. (3 marks for all four; 2 marks for three; 1 mark for two)

AI 2. AI and Its Applications

1. a) chest X-rays
2. a) classes
3. a) count wildlife
4. create classes, add examples, train, test with new examples (2 marks; 1 mark if one step is out of place)
5. Its data had no variety. Add photos with different backgrounds and light, then retrain. (1 mark for each)
6. Classes: ripe, unripe. Data: 20-30 photos of each, with different light, backgrounds and angles. Test: new tomatoes the model has not seen. (1 mark for each)

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AI 3. Data and Fairness in AI

1. b) under-represented
2. a) representative of every group
3. a) each group separately
4. It may fail more often on dark-skinned faces (under-represented). Add more photos of dark-skinned faces to rebalance, then test again. (1 mark for the problem, 1 mark for the fix)
5. Its training data had many more sentences about male doctors (over-represented) (2 marks; 1 mark for 'the data')
6. Boys are over-represented, so the model copies their choice. Collect more answers from girls and check each group separately. It should not use gender to decide. (1 mark for each)

AI 4. Ethics and Responsible AI

1. a) consent
2. a) deny it
3. a) people
4. For example: look for it on trusted news sites; check who posted it and when (1 mark for each)
5. 'Only while using the app', because it needs your location only when you are finding a route (1 mark for each)
6. Transparent: tell people why they were refused. Fair: check it treats every group equally. Accountable: a person reviews and is responsible for the final decision. (1 mark for each)

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Class record sheet

Class 8 ____ Section ____ Term ____. Write each score out of 10. You may photocopy this page.

Student	1	2	3	4	5	6	7	AI1	AI2	AI3	AI4	Avg

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